Chapter 34 — Deployment Topologies

Field	Value
Status	Revised after whole-book constitutional and cross-chapter audit
Version	0.2
Scope	Define deployable HAL topologies that preserve one active constitutional control plane while allowing distributed, replaceable execution and multiple Presences.
Constitutional basis	Decisions 1–2, 17–24, 29, 37–43, 45, 47, 51
Owner review	No unresolved Owner decision after whole-book audit

1. Purpose

Define deployable HAL topologies that preserve one active constitutional control plane while allowing distributed, replaceable execution and multiple Presences.

2. Authoritative Responsibilities

Component	Authoritative responsibility
Deployment Controller	Topology manifest, environment, service placement, versions, dependencies, and desired state
HAL Core Runtime	Constitutional kernel, governance, identity, authority, policy, canonical coordination, and audit
HAL Node Runtime	Authenticated capability execution, local health, bounded cache, evidence, and restricted offline mode
Presence Runtime	Device interaction, sensors/outputs, privacy context, and non-authoritative local state

3. Core State and Records

Object	Required content
Deployment Manifest	Environment, hosts, services, identities, networks, storage, policies, versions, and recovery profile.
Core Host Profile	Dedicated Mac mini or successor, kernel services, canonical storage, security, observability, and reserve.
Node Profile	GX10 or future compute node identity, capabilities, resource limits, software, health, and delegation.
Environment Boundary	Simulation, test, shadow, canary, production, recovery, or emergency with permitted effects.

4. Runtime Workflow

1. Validate deployment manifest, host identities, signed artifacts, configuration, secrets, and network boundaries.
2. Bootstrap the dedicated HAL Core control plane and establish the constitutional lease; a recovery successor lease is capped at 24 hours and every extension requires fresh Owner authorization.
3. Restore or initialize Constitution, Owner identity, authority, policy, audit, ledgers, and canonical state.
4. Join compute nodes through the controlled recovery/join pipeline and advertise verified capabilities.
5. Start replaceable services and Presences only after their dependencies and authority boundaries are ready.
6. Route execution dynamically while governance and canonical state remain controlled by HAL Core.
7. Continuously reconcile desired and observed topology and test recovery into an isolated environment.

5. Interfaces and Contracts

Commands, queries, events, records, and evidence are typed, versioned, authenticated, authorized, correlated, and classified. Mutation is accepted only by the authoritative owner. Every external or reality-changing boundary carries identity, policy, authority, provenance, integrity, time, and transaction context.

6. Failure and Recovery

- Core unavailable: nodes retain only bounded delegated work; no node becomes an equal independent HAL.

- Node unavailable: reassign compatible execution or disclose degraded capability.
- Environment boundary uncertain: block real-world effects until production identity is proven.
- Topology drift: reconcile ordinary changes; escalate identity, policy, or constitutional divergence.

7. Constitutional Guarantees

- The reference household topology is one dedicated HAL Core with lightweight node and Presence runtimes.
- Distributed execution never distributes constitutional ownership.
- Topology and hardware may evolve without changing HAL identity or capability contracts.
- Simulation, test, recovery, and production deployments cannot accidentally share reality-changing authority.

8. Prohibited Behaviors

- No component may silently broaden or narrow constitutional authority, reinterpret Owner intent, erase dissent, or promote derived state to authority.
- No failure, degraded dependency, emergency, or optimization may silently change identity, policy, trust, authentication, evidence, or verification rules.
- No model, service, provider, node, Presence, credential, relationship, environment, or network location receives constitutional authority by implication.

9. Observability and Verification

The subsystem records identity, correlation, authority/policy context, inputs, state versions, decisions, limitations, failures, recovery, and outcomes. Verification is risk-scaled, reproducible, and stored as durable evidence. Status and explanation derive from governed state rather than model assertion.

10. Source Alignment and Review

This chapter implements Decisions 1–2, 17–24, 29, 37–43, 45, 47, 51. Final source-alignment and constitutional disposition is recorded in the batch audit reports; embedded approval text is not treated as review evidence.

Whole-Book Audit Resolution

The following requirements supersede any ambiguous earlier wording in this chapter and are incorporated into this edition:

- The Mac mini is the normal Primary Constitutional Host, not an irreplaceable identity anchor.
- Only one active constitutional control plane may hold a valid lease.
- A pre-registered successor may hold a verified recovery lease for no more than 24 hours; extensions require fresh Owner authorization and permanent reassignment requires the Owner Authorization Ceremony.